Road Damage Detection with YOLO

Training & Evaluation Report — RDD2022 (7 countries, 4 classes)

Model: Ultralytics YOLO11s · Input 512 px · 50 epochs · NVIDIA RTX 4060 (8 GB)

Generated: 2026-11-11

Headline results (held-out TEST split, 5,758 images)

0.613	0.330	0.647	0.578
mAP@0.50	mAP@0.50:0.95	Precision	Recall
all classes	all classes	all classes	all classes

This report documents an end-to-end object-detection pipeline that locates four types of road damage — longitudinal cracks, transverse cracks, alligator cracks and potholes — in street-level photographs from the multi-national RDD2022 dataset. A YOLO11s network was trained from COCO-pretrained weights on 26,869 images spanning seven countries and evaluated on a held-out test set of 5,758 images, with a per-class and per-country breakdown. The sections below describe the data, the training configuration, the quantitative results and the main take-aways.

Deliverables

weights/best.pt · reports/RDD2022_training_report.pdf (this file)

reports/metrics_test.md · reports_val/metrics.md · runs/detect/rdd_yolo/ (curves, confusion matrix)

runs/predict/ (annotated sample images) · reports/*.png (dataset figures)

1. Dataset

RDD2022 (Road Damage Dataset 2022, released for the CRDDC'2022 challenge) is a multi-national collection of road images annotated with four damage types scored by the challenge: longitudinal cracks (D00), transverse cracks (D10), alligator cracks (D20) and potholes (D40). The archive used here was already converted to the YOLO format and split into train/val/test. It carried a fifth, non-challenge class (a repair-type code); those 6,544 boxes were dropped so the data matches the four official classes. Images without any of the four damages are kept as background examples, which reduces false positives on clean asphalt.

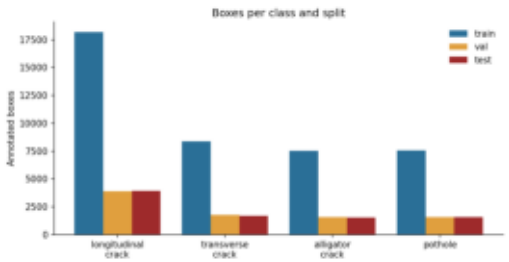
Images and annotations per split

Split	Images	With damage	Boxes
Train	26,869	17,902	41,668
Val	5,758	3,743	8,776
Test	5,758	3,785	8,724
Total	**38,385**	**25,430**	**59,168**

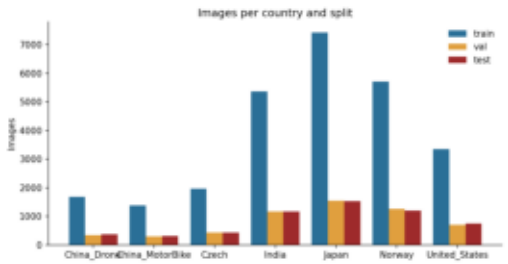
Boxes per class (all splits)

Class	Boxes
Longitudinal crack (D00)	26,016
Transverse crack (D10)	11,830
Alligator crack (D20)	10,617
Pothole (D40)	10,705

Class distribution



Images per country



2. Training setup

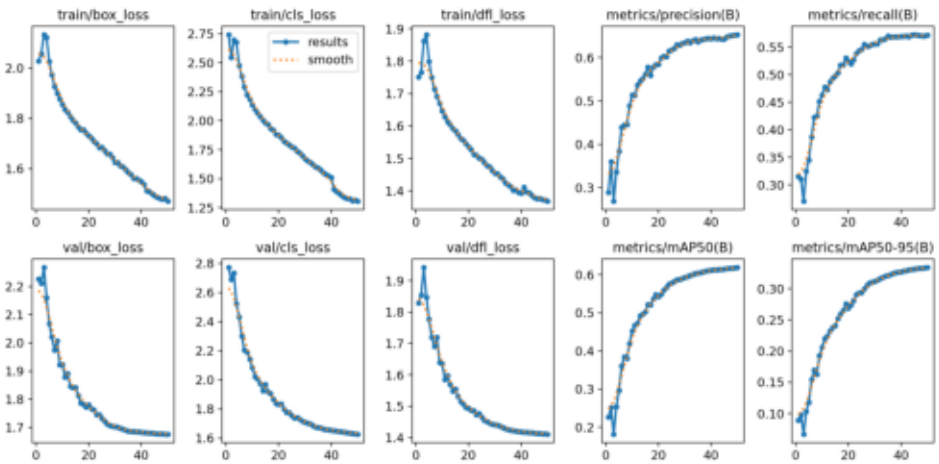
The detector was fine-tuned from the COCO-pretrained YOLO11s checkpoint. Training ran on a single NVIDIA GeForce RTX 4060 Laptop GPU (8 GB) using CUDA, in mixed precision. A fixed batch size of 8 was chosen to leave head-room on the 8 GB card; this also keeps the run robust when the GPU is shared with other applications.

Setting	Value
Architecture	YOLO11s (9.4 M parameters)
Initial weights	COCO-pretrained (yolo11s.pt)
Image size	512 x 512
Epochs	50 (early-stopping patience 25)
Batch size	8 (fixed)
Optimizer	auto (SGD-family), cosine LR schedule
Augmentation	mosaic; rotation +/-5 deg, shear 2 deg; h-flip 0.5; no v-flip
Hardware	NVIDIA RTX 4060 Laptop, 8 GB, CUDA 12.4, torch 2.6
Wall-clock time	7.67 hours (50 epochs)

Augmentation is deliberately conservative on geometry: crack orientation is what separates a longitudinal from a transverse crack, so rotation and shear are kept small and vertical flips are disabled (they would also break the road's perspective).

Training curves

Losses and validation metrics across the 50 epochs (validation mAP50 rose steadily to ~0.62 and plateaued)



3. Results — overall and per class

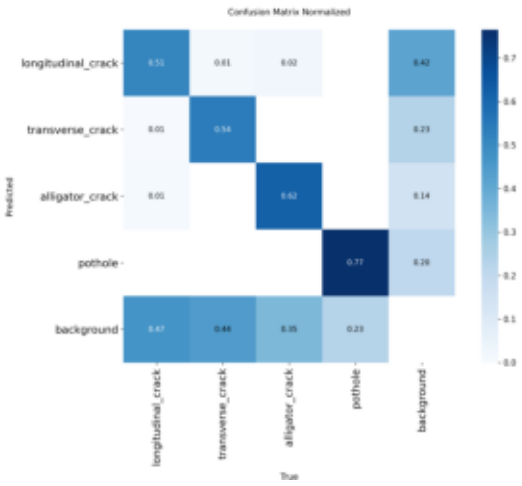
All numbers below are computed on the held-out TEST split (5,758 images), which the model never saw during training. The validation split gives nearly identical overall scores (val mAP50 = 0.619 vs test 0.613), indicating the model generalises and is not over-fitted.

Class	AP50	AP50-95	Prec.	Recall
Longitudinal crack (D00)	0.535	0.282	0.629	0.500
Transverse crack (D10)	0.532	0.245	0.588	0.518
Alligator crack (D20)	0.647	0.340	0.693	0.572
Pothole (D40)	0.737	0.451	0.678	0.721
All classes (mean)	**0.613**	**0.330**	**0.647**	**0.578**

- Potholes are detected best (AP50 0.737): they are compact, high-contrast objects.
- Alligator cracking (AP50 0.647) is also strong — it has a distinctive textured pattern.
- The two thin linear cracks (longitudinal / transverse, AP50 ~0.53) are hardest: they are low-contrast, elongated and easy to confuse with each other, especially after down-scaling to 512 px.

Confusion matrix (normalised)

Rows = predictions, columns = ground truth. Most confusion is between the two linear-crack classes and the background.



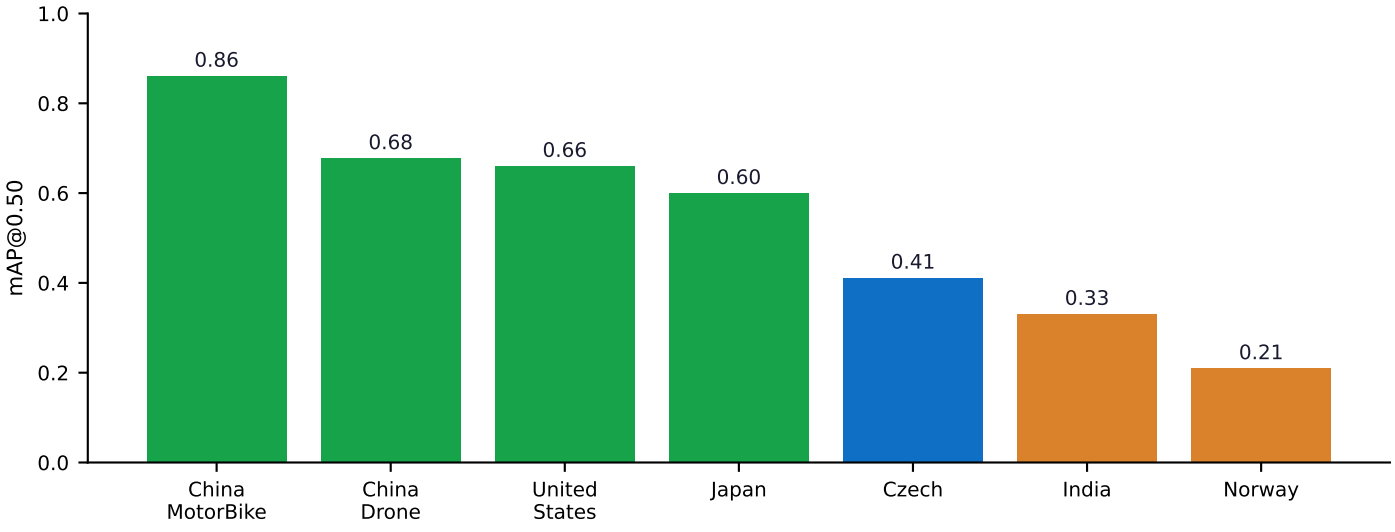
4. Results — per country

Because RDD2022 pools seven countries with different cameras, road types and image resolutions, per-country performance varies widely. The table is sorted by mAP50.

Country	Images	mAP50	mAP50-95	Prec.	Recall
China MotorBike	305	0.862	0.578	0.852	0.790
China Drone	377	0.678	0.402	0.738	0.601
United States	753	0.661	0.363	0.659	0.619
Japan	1,524	0.600	0.303	0.608	0.591
Czech	436	0.412	0.152	0.480	0.400
India	1,166	0.331	0.158	0.423	0.380
Norway	1,197	0.210	0.087	0.331	0.248

- China_MotorBike (0.862) and China_Drone (0.678) are strongest — clean, close-range imagery.
- United_States (0.661) and Japan (0.600) are solid and well represented in training.
- Czech (0.412) and India (0.331) are weaker: India in particular has cluttered scenes and many small/ambiguous cracks.
- Norway (0.210) is the weakest by far. Its native images are 3650x2044; down-scaling them to 512 px erases thin cracks. Training Norway at a larger image size (and more VRAM) is the recommended fix.

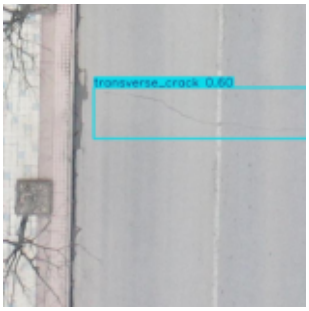
Per-country mAP@0.50



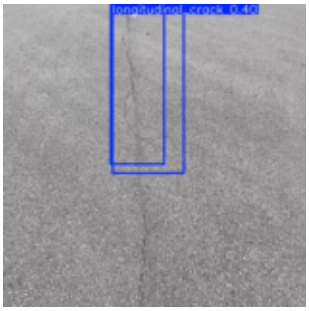
5. Qualitative predictions

The model was run on one held-out test image per country (confidence >= 0.25). Boxes and labels below are the model's predictions.

China (Drone)



China (MotorBike)



United States



Japan



India



Norway



6. Conclusions and next steps

A YOLO11s detector trained on the full seven-country RDD2022 set reaches mAP@0.50 = 0.613 and mAP@0.50:0.95 = 0.330 on a held-out test set of 5,758 images. Performance is best on compact, high-contrast damage (potholes, alligator cracking) and on clean imagery (the two China subsets, the US and Japan); it is weakest on thin linear cracks and on Norway's high-resolution images, which lose detail when down-scaled to 512 px. Train/val/test scores agree closely, so the model generalises well.

How to run the trained model

```
python -m rdd predict --source path/to/photo.jpg --weights weights/best.pt --imgsz 512
```

Recommended improvements

- Train at 640 px (or higher) on a larger GPU — the single biggest lever for the thin-crack and Norway cases; a bigger batch would also speed things up.
- Use a larger backbone (YOLO11m/l) once more VRAM is available for a few extra mAP points.
- Handle Norway separately at a higher resolution, or exclude it when reporting the headline number.
- Add more training epochs (100+) with the cosine schedule; the curve had not fully saturated.
- Address class imbalance for transverse cracks (the rarest linear class) with targeted sampling.

Method notes: the four classes are the official challenge codes (D00/D10/D20/D40); a fifth repair-type code present in the source archive was dropped. Damage-free images are retained as background. Per-class metrics are mapped through the detector's class index so an absent class in one country cannot shift another class's numbers.

Dataset: RDD2022 (CC BY-SA 4.0). Model: Ultralytics YOLO (AGPL-3.0). Report auto-generated from reports/metrics_test.json.